

2.11

$$I_{D2} = \frac{V_0}{R_{D2} \parallel (R_F + R_S)} = 0.258 \text{ mA} \quad V_{eq1} = V_{DD} \frac{R_{G2}}{R_{G1} + R_{G2}} = 4 \text{ V}$$

$$R_{eq1} = R_{G1} \parallel R_{G2} = 60 \text{ k}\Omega$$

$$\begin{cases} I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_T}\right)^2 \\ V_{GS} = V_{eq1} - I_D R_S \end{cases} \rightarrow \frac{V_{eq1}}{R_S I_{DSS}} - \frac{V_{GS1}}{R_S I_{DSS}} = 1 + \frac{V_{GS1}^2}{V_T^2} - \frac{2V_{GS1}}{V_T} \rightarrow \frac{V_{GS1}^2}{0.25} - \frac{2V_{GS1}}{-0.5833} + \frac{1 - \frac{V_{eq1}}{R_S I_{DSS}}}{+0.567} = 0$$

$$V_{GS1} = 3.174 \text{ V} > V_{T1} = 2 \text{ V}$$

$$I_{D2} = 0.258 \text{ mA}$$

$$I_{D1} = 0.826 \text{ mA}$$

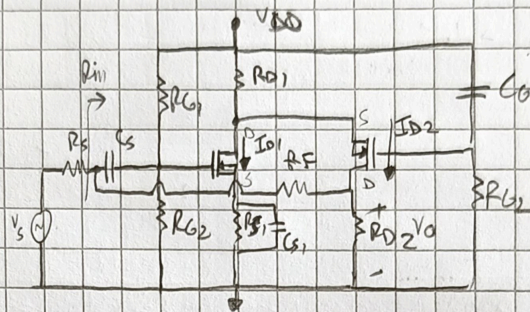
$$\frac{I_D}{I_{DSS}} = 1 + \frac{V_{GS}^2}{V_T^2} - \frac{2V_{GS}}{V_T} \rightarrow \frac{V_{GS2}^2}{V_T^2} - \frac{2V_{GS2}}{V_T} + 1 - \frac{I_{D2}}{I_{DSS}} = 0$$

$$V_{GS2} = -3.804 \text{ V} < V_T = -3 \text{ V}$$

$$V_{GS1} = -V_{GS2} - I_D R_{S1} = 2.98 \text{ V}$$

$$V_{GS2} = -(-V_{GS2} - V_0) = V_{GS2} + V_0 = -1.8 \text{ V}$$

$$R_{D1} = \frac{V_{DD} + V_{GS2}}{I_{D1} + I_{D2}} = 5.716 \text{ k}\Omega$$



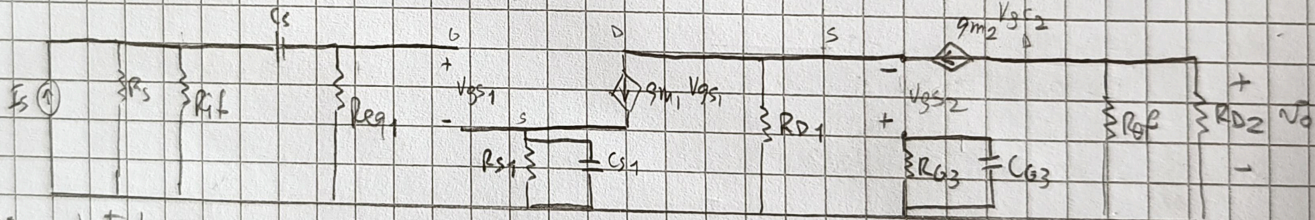
Topology: current mixing - voltage sampling
 Norton eq.
 (Shunt) (Shunt)
 $r_o = \text{transresistance amplifier}$

$$\begin{aligned} R_{if} &= R_F \\ R_{of} &= R_F \\ \beta &= -\frac{1}{R_F} \end{aligned}$$

$$g_{m1} = \frac{2I_{D1}}{V_{GS1} - V_{T1}} = 1.41 \cdot 10^{-3} \text{ S}$$

$$g_{m2} = \frac{2I_{D2}}{V_{GS2} - V_{T2}} = 6.618 \cdot 10^{-4} \text{ S}$$

$$I_S = \frac{V_S}{R_S}$$



with C_S, C_1, C_3 shorted

$$A_{v1} = \alpha_{CG} \cdot \alpha_{CS} \cdot \alpha_i = \frac{v_o}{v_i} \quad \beta \approx 0$$

$$\alpha_{CG} = \frac{+g_{m2} (R_{of} \parallel R_{D2})}{1} = 6.814 \text{ [V/V]}$$

$$\alpha_{CS} = \frac{1}{g_{m2}} = 1558 \Omega$$

$$\alpha_i = -g_{m1} (R_{D1} \parallel \frac{1}{g_{m2}}) = -1.726 \text{ [V/V]}$$

$$\alpha_i = R_S \parallel R_{if} \parallel R_{eq1} = 923 \Omega$$

$$A_v = \alpha_i \alpha_{CS} \alpha_{CG} = -7.669 \text{ [K]} \quad A_v = -5.075 \text{ [V/V]}$$

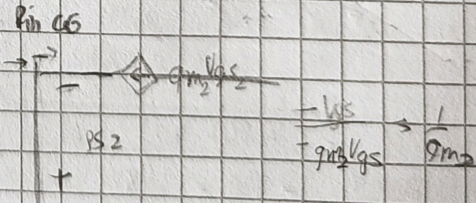
$$A_F = \frac{A_v}{1 + \beta A_v} = -5.075 \text{ k}\Omega = \frac{V_o}{V_i}$$

$$A_v = \frac{v_o}{v_i} = \frac{A_F}{R_S} = -5.075 \text{ [V/V]}$$

$$R_{in}^{ac} = \alpha_i = 923 \Omega$$

$$R_{in}^F = \frac{R_{in}^{ac}}{1 + \beta A_v} = 611 \Omega$$

$$R_{in} = \frac{1}{\frac{1}{R_{in}^F} + \frac{1}{R_S}} = 1.571 \text{ k}\Omega$$



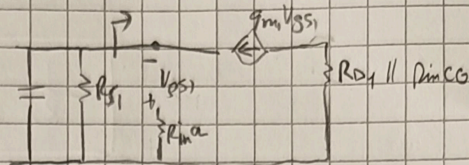
short circuit time constant method (SCTC)

$$P_L = \frac{1}{g_m} \left(\frac{1}{C_{gs} R_{gs}^{sc}} + \frac{1}{C_{gs} R_{gs}^{sc}} + \frac{1}{C_{gs} R_{gs}^{sc}} \right)$$

$R_{gs}^{sc} = R_s + R_{in} \rightarrow$ no need of feedback theory because referring to starting circuit

$R_{g3}^{sc} = R_{g3} \rightarrow$ not influenced by feedback

$R_{s1}^{sc} = R_{s1} \parallel \frac{1}{g_{m1}} \rightarrow$ feedback theory to be applied



$R_{g3} \gg R_{gs}^{sc} > R_{s1}^{sc}$

↑
not relevant contribute

$R_{gs}^{sc} = 2571 \Omega$

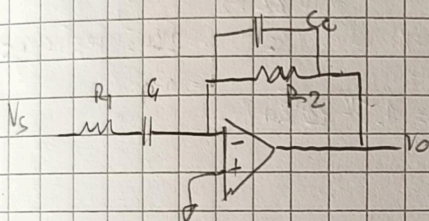
$R_{g3}^{sc} = 100k\Omega$

$R_{s1}^{sc} = 415 \Omega$

$f_{L1}^a = \frac{1}{2\pi C_{gs} R_{gs}^{sc}} = 38 \text{ Hz} \rightarrow f_{L1}^F = \frac{f_{L1}^a}{1 + A_{OL}\beta} = 254 \text{ Hz}$

$f_L = \frac{1}{\frac{2\pi C_{gs} R_{gs}^{sc}}{6.19 \text{ Hz}} + \frac{f_{L1}^F}{254 \text{ Hz}} + \frac{1}{\frac{2\pi C_{g3} R_{g3}^{sc}}{0.16 \text{ Hz}}}} = 32 \text{ Hz}$

Ex2



$A(s) = \frac{A_{db}}{(1+s/\omega_0)(1+s/\omega_1)}$

$R_1 = 10k\Omega$

$R_2 = 220k\Omega$

$C_1 = 100 \text{ nF}$

$A_{db} = 10^5$

$\omega_0 = 10 \text{ rad/s}$

$\omega_1 = 5000 \text{ rad/s}$

1) $A_V(s) = \frac{V_o}{V_s} = -\frac{Z_2}{Z_1} = -\frac{R_2}{R_1 + \frac{1}{sC_1}} = -\frac{sGR_2}{1+sGR_1}$

2) $T = -\frac{V_r}{V_s} = A_V(s) \cdot \frac{Z_1}{Z_1 + Z_2}$

$\beta = \frac{R_1 + \frac{1}{sC_1}}{R_1 + \frac{1}{sC_1} + R_2} = \frac{1+sGR_1}{1+sG(R_1+R_2)}$

$\beta_0 = 1$

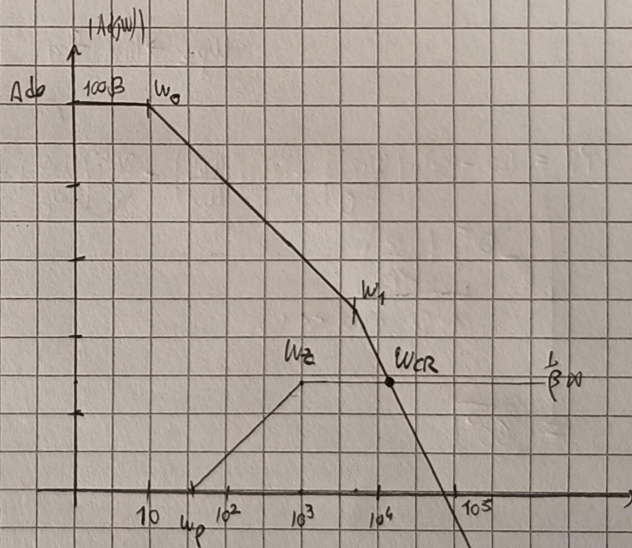
$\beta_{\infty} = \frac{R_1}{R_1+R_2}$

$\beta_{db} = 1$

$\frac{1}{\beta_{db}} = 1 + \frac{R_2}{R_1} = 23 = 27 \text{ dB}$

$\omega_2 = \frac{1}{C_1 R_1} = 10^3 \text{ rad/s}$

$\omega_P = \frac{1}{G(R_1+R_2)} = 43.5 \text{ rad/s}$



$A_{db}\omega_0 = \omega_1 A_X$

$A_X \omega_1^2 = \omega_{cr}^2 \frac{1}{\beta_{db}}$

$A_X = \frac{A_{db}\omega_0}{\omega_1}$

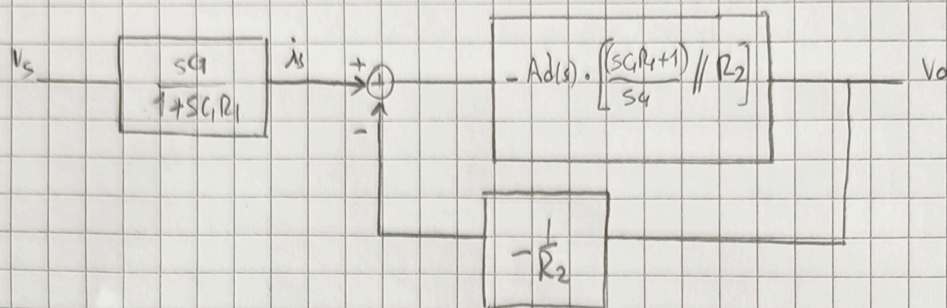
$\beta_{db} A_{db} \omega_0 \omega_1 = \omega_{cr} = 1.47 \cdot 10^4 \text{ rad/s}$

$PM = 180 - 2 \arctan\left(\frac{\omega_{cr}}{\omega_0}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_1}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_P}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_2}\right)$

$= 15.1^\circ$ low phase margin

3) inverting configuration: current mirroring - voltage sampling

$$\frac{1}{Z_1} = \frac{1}{R_1 + \frac{1}{sC_1}} = \frac{sC_1}{1 + sC_1 R_1}$$



$$4) K_S(s) = \frac{sC_1}{1 + sC_1 R_1}$$

max value of C_2 $C_2 \ll C_1 \rightarrow \omega_{p2} = \frac{1}{C_2(R_2 \parallel R_1)} \rightarrow \omega_{p2} > \omega_{CR}$

high frequency pole

$$\frac{1}{C_2(R_2 \parallel R_1)} > \omega_{CR}$$

idn't want the pole but also the zero close to ω_{CR}

$$C_2 < \frac{1}{\omega_{CR} R_2 \parallel R_1} = 7.1 \text{ nF} \ll C_1 = 100 \text{ nF}$$

$$\omega_{z2} = \frac{1}{C_2 R_2} \rightarrow \omega_{z2} > \omega_{CR} \rightarrow C_2 < \frac{1}{\omega_{CR} R_2} = 309 \text{ pF}$$

more restrictive condition

$C_2 \ll C_1$
High Freq

low freq

$$\omega_{z1} = \frac{1}{C_1 R_1} = 10^3 \text{ rad/s}$$

$$\omega_{z2} = \frac{1}{C_2 R_2} = \omega_{CR} \text{ high frequency}$$

$$\omega_{p1} = \omega_p \approx 43.5 \text{ rad/s} (C_2 \ll C_1)$$

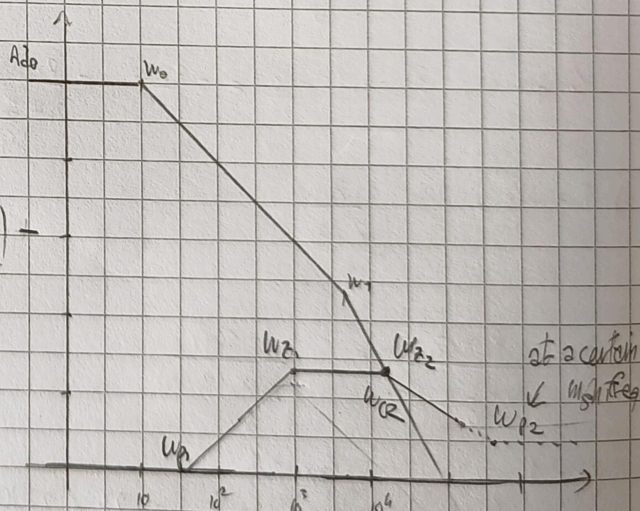
$\omega_{p2} \rightarrow \text{high freq}$

$$PSI = 180 - \arg\left(\frac{\omega_{CR}}{\omega_0}\right) - \arg\left(\frac{\omega_{CR}}{\omega_{z1}}\right) - \arg\left(\frac{\omega_{CR}}{\omega_{p1}}\right) + \arg\left(\frac{\omega_{CR}}{\omega_{z1}}\right) + \arg\left(\frac{\omega_{CR}}{\omega_{z2}}\right) -$$

$$- \arg\left(\frac{\omega_{CR}}{\omega_{p2}}\right)$$

$\omega_0, \omega_{CR} \ll \omega_{p2}$

$$= 60^\circ$$



1 ora e 35 minuti